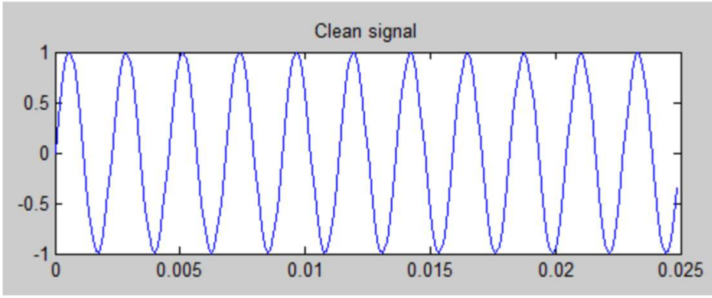
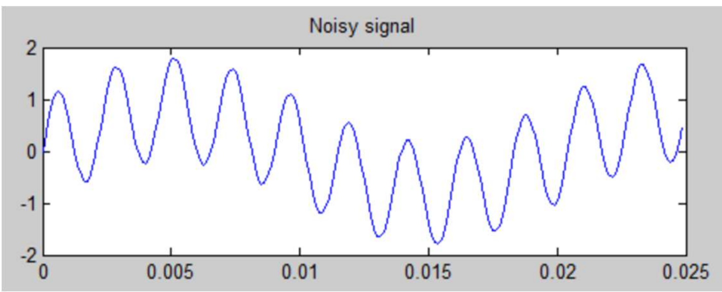
AUDIO NOTCH FILTER PROJECT FIGURES

Figure 1: Clean Signal



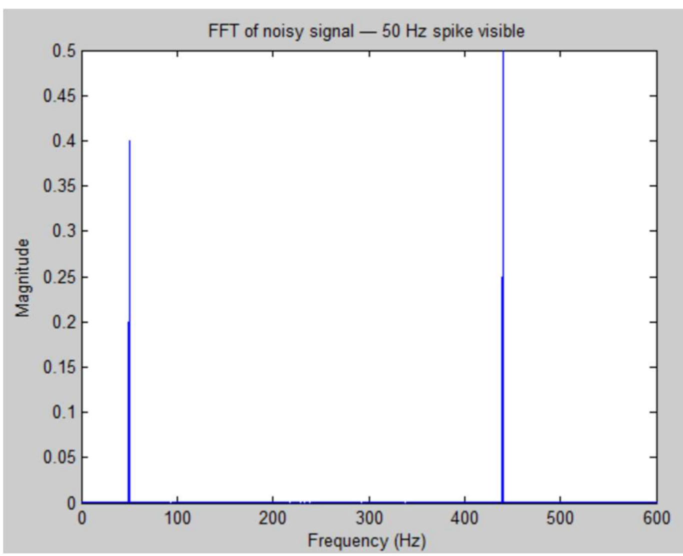
Time-domain plot of the original 440 Hz sine wave with amplitude ±1, representing the clean signal before any interference is introduced.

Figure 2: Noisy Signal



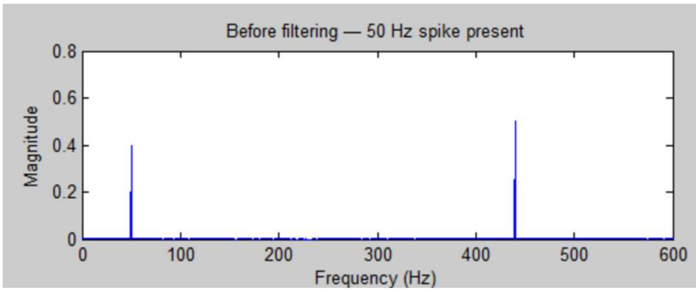
Time-domain plot of the corrupted signal after 50 Hz power-line noise is added, causing the amplitude to swing between ±2 and visibly distorting the waveform shape.

Figure 3: FFT Noise Spike



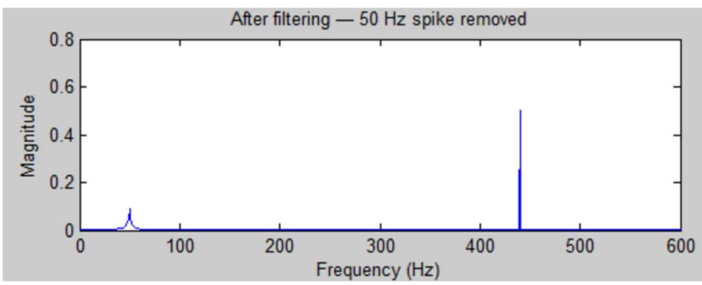
FFT magnitude spectrum of the noisy signal revealing two distinct frequency components, the 50 Hz noise spike (magnitude ~0.4) and the 440 Hz audio tone (magnitude ~0.5) both clearly isolated against a flat baseline.

Figure 4: FFT Before



FFT spectrum of the noisy signal before filtering. Both the 50 Hz power-line spike and 440 Hz audio tone are clearly present, establishing the baseline for comparison.

Figure 5: FFT Money Shot



FFT spectrum after applying the notch filter. The 50 Hz spike is significantly attenuated while the 440 Hz signal is fully preserved, confirming the filter works as intended.